



Regular article

A two-step procedure for the design of perfusion bioreactors

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HIGHLIGHTS

- 2-step procedure for the design and development of CHO cell perfusion cultures.
- Step 1: Definition of suitable minimum CSPR value at constant P or constant VCD.
- Step 2: Process optimization at constant CSPR but elevated values of VCD and P.
- Procedure enables high product yield and improved process performance.
- Product quality at steady state only shows minor variations.

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ABSTRACT

A two-step procedure for the design and optimization of mammalian cell perfusion cultures for therapeutic protein production, for a given expression system, is developed. The considered operating variables include the cell specific perfusion rate (CSPR), the perfusion rate (P) and the viable cell density (VCD). The procedure is articulated in two steps, each including a reactor operation through which different steady states are explored sequentially. In the first one, a suitable minimum value of the CSPR is found. This can be achieved through two alternative approaches. These include keeping the perfusion rate or the VCD constant, while changing the other so as to decrease the CSPR. Cellular growth and metabolite in- and output rates decrease when minimizing the CSPR resulting in a lower bleed rate and a better process performance. In the second step, steady states at constant CSPR but increasing VCD values are investigated, by properly adjusting the perfusion rate. In this case, the cellular activity remains constant, while the harvest yield and volumetric productivity further increase with higher perfusion rates. The product quality (charge variants, N-linked glycosylation) investigated during both steps of the procedure shows only minor changes among the different steady states. If product quality properties obtained at this point of the procedure are not satisfactory, they have to be fine-tuned in an additional step. As reported previously, the addition of supplements in the feed stream can be applied to induce significant changes in product quality.

1. Introduction

The biopharmaceutical industry has grown rapidly over the last thirty years since the first commercialization of recombinant proteins in the 1980s [1,2]. The market of monoclonal antibody (mAb) products is expected to further grow in the near future with a combined world-wide sale of above \$ 125 billion in 2020 [1,3]. While demand for manufacturing capacity remains high [4,5], continuous perfusion cell cultures have regained attention as production systems [6,7]. Perfusion bioreactors can provide higher daily productivities, a smaller facility footprint, more homogeneous product quality due to shorter residence

time, and steady-state operation. Further, they enable the integrated continuous end-to-end production of the expressed protein [5,8]. Several studies have proven the feasibility of the end-to-end integration of continuous up- and downstream processes [9–12], which in some situations are considered as the most convenient manufacturing strategy. Besides, perfused bioreactors have emerged as a valuable tool for intensification of current batch-wise technologies [13–19].

Perfusion processes have been studied since the 1990s when first investigations on suitable control strategies, real-time monitoring, and process stability were performed [20–26]. The ongoing improvements in cell lines, medium formulations and equipment facilitate today's high

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Nomenclature

AA	Amino acid
AMM	Ammonia
ALA	Alanine
ASN	Asparagine
ATF	Alternated tangential flow
CHO	Chinese hamster ovary
CSPR	Cell specific perfusion rate
CV	Charge variant
DOT	Dissolved oxygen tension
GalT	Galactosyltransferase

GLC	Glucose
GS	Glutamine synthetase
HF	Hollow fiber
LAC	Lactate
mAb	Monoclonal antibody
MSX	Methionine sulfoximine
P	Perfusion rate
PES	Polyether sulfone
PHE	Phenylalanine
RV	Reactor volume
SER	Serine
VCD	Viable cell density

cell density perfusion cultures [27–29]. However, medium consumption and associated costs, as well as operational complexity and labor intensity still represent major disadvantages [5,30].

The development of a perfusion bioreactor process consists of several phases, starting from media design and clone selection. Once those are fixed, we need to design a process at the laboratory scale that subsequently can be scaled-up and operated at clinical and commercial scales. This implies to define the reactor volume specific flowrate of the perfusion stream, P (RV/day), and the viable cell density (VCD), X_V , (cells/mL) in order to realize a stable perfusion culture while “optimizing” its performance. This is in general quantified in terms of maximum productivity, defined as amount of protein produced per reactor volume and time, and minimum medium consumption, defined as the amount of medium consumed per mass of protein produced. The characterization of the bioreactor steady state is completed by the titer and the reactor volume specific bleed flowrate, B through the corresponding mass balances:

$$C_{\text{Titer}} = \frac{X_V q_{\text{mAb}}}{P} \quad (1)$$

$$B = \mu_{\text{specific}} \quad (2)$$

where q_{mAb} (e.g., pgProtein/cell/day) represents the cell specific productivity and μ_{specific} (e.g., 1/day) is the cell specific growth rate.

The design of the bioreactor operating conditions has to satisfy certain constraints coming from the adopted set-up and expression system [31,32]. Typically, the VCD cannot exceed a maximum value due to limitations in oxygen supply or mixing (viscosity). The perfusion rate P is upper-bounded by the capacity of the cell filtration unit and lower-bounded by the average residence time value of the protein. This last constraint is particularly important for unstable products.

A particularly significant parameter often used in the operation of perfusion bioreactors is the cell specific perfusion rate (CSPR), (pL/cell/day), defined as the amount of medium fed to a single cell per day [31,32]:

$$\text{CSPR} = \frac{P}{X_V} \quad (3)$$

It can be shown that by decreasing the CSPR it is possible to increase productivity while decreasing medium consumption and therefore improve process performance. Of course, for a given medium, there is a critical value of CSPR below which the culture cannot be sustained and therefore it was concluded that the optimal performance of the bioreactor is obtained at the minimum value of CSPR at which the perfusion culture is stable, referred to as CSPR_{min} [31]. This parameter is strongly correlated to the medium depth, that is the maximum VCD a given medium can support for a given perfusion rate [33,34].

In principle, two approaches can be envisioned to estimate the CSPR_{min} . The *push-to-low* approach, originally proposed by Konstantinov et al. [31], consists of a stepwise reduction of the perfusion rate at a fixed viable cell density so as to explore decreasing values of the CSPR. Alternatively, the *push-to-high*

approach involves a stepwise increase of the viable cell density at a constant perfusion rate. In both cases the CSPR decreases at each step until either medium limitations or metabolite inhibitions result in process instability (e.g., cell death or decrease in cell specific productivity), which in principle occurs at the same CSPR value corresponding to the CSPR_{min} . However, care must be taken because the path followed to reach the CSPR_{min} value is not the same. In the first one, the perfusion rate decreases, which may lead to excessive average residence times in the case of fragile products, in the second the VCD increases, which may lead to unfeasibly large values. The two procedures are illustrated schematically in Fig. 1.

Once the CSPR_{min} has been identified, we are left with one operating parameter that can be selected to further improve the bioreactor performance that is P or VCD: their ratio being fixed. This second design step includes a stepwise increase of both the perfusion rate and the viable cell density in order to keep the CSPR constant. It is clear that this can be done up to a certain extent, defined by the given set-up, due to the occurrence in limitations of oxygen transfer, carbon dioxide removal, viscosity or others. It is worth noting that, in the case of fragile proteins this second step can be highly beneficial, as the increased flow rates shorten the protein's residence time inside the reactor, and therefore prevent undesired degradation.

A particularly useful approach to the design of perfusion bioreactors is to investigate sequentially various sets of operating conditions by reaching, for each given set, steady-state operation. For this, the reactor is operated for more than 7 days before changing to the next set of operating conditions. This has been demonstrated to be a reasonable estimate of the transient behavior needed to reach steady state [35,36].

In this work, we illustrate the stepwise sequential strategy described above for the design of a perfusion bioreactor for a given cell line and medium composition. The first step evaluates a suitable minimum CSPR value, CSPR_{min} within three sequential steady states by applying both

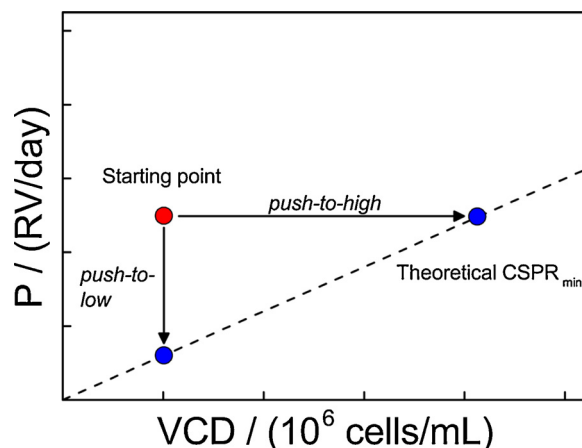


Fig. 1. Schematic representation of the push-to-low and the push-to-high approach to determine CSPR_{min} .

the *push-to-high* and the *push-to-low* strategy. In the second step, we further improve the reactor performance for one of the tested CSPRs by investigating sequentially several pairs of P and VCD values while keeping constant their ratio, the CSPR. In summary, by combining these two steps one can identify the operating conditions that optimize the bioreactor performance in terms of productivity and medium utilization.

2. Material and methods

2.1. Cell culture

A mAb producing proprietary CHO cell line was cultured in chemically defined medium (Merck Serono SA, Switzerland). Cells were thawed and expanded in cell culture medium containing 25 μ M methionine sulfoximine (MSX). The expansion was performed at 36.5 °C, 5% CO₂, and 90% humidity for one week in a shaking incubator (Adolf Kuhner AG, Switzerland). Cells were passaged every two days.

2.1.1. Seed bioreactor

In a next step, cells were incubated in a perfused seed bioreactor (DASGIP®, Eppendorf AG, Switzerland, 2.5 L glass vessel, working volume 2 L). The system employed an external loop including a polyether sulfone (PES) hollow fiber (HF) module of 25 cm length, 0.5 μ m pore

size, and a filtration area of 1570 cm² (Spectrum Laboratories, Breda, the Netherlands) for cell retention driven by a bearingless centrifugal pump (PuraLev® 200MU Levitronix AG, Switzerland). The pump module was operated and controlled at 1.6 L/min. The VCD was increased from 0.3 to 55 $\times 10^6$ cells/mL during seven days as shown in **Figure S1** in the supplementary material. The reactor weight was kept constant by gravimetric feed control. The harvest rate was set to 1 RV/day during the first four days, and then, increased to 2 RV/day to prevent nutrient limitations.

2.1.2. Production bioreactors

Cells were transferred to a previously developed and well characterized perfusion bioreactor system which was operated at a working volume of 1.5 L. The reactor was connected to an external ATF2 device (Repligen Corporation, Waltham, MA) using the same PES HF type for cell retention. The ATF device was operated at a constant exchange flow rate of 1.2 L/min. The bioreactor was operated in terms of pH, DO, agitation and temperature set points as described in previous publications [28].

Three experimental runs were carried out. In each one, two or three different steady states (SS1, SS2, and SS3), separated by a transient of at least 9 days were investigated sequentially. The main investigated parameters were P (RV/day), VCD (cells/mL), and CSPR (pL/cell/day) as defined in Eq. (1).

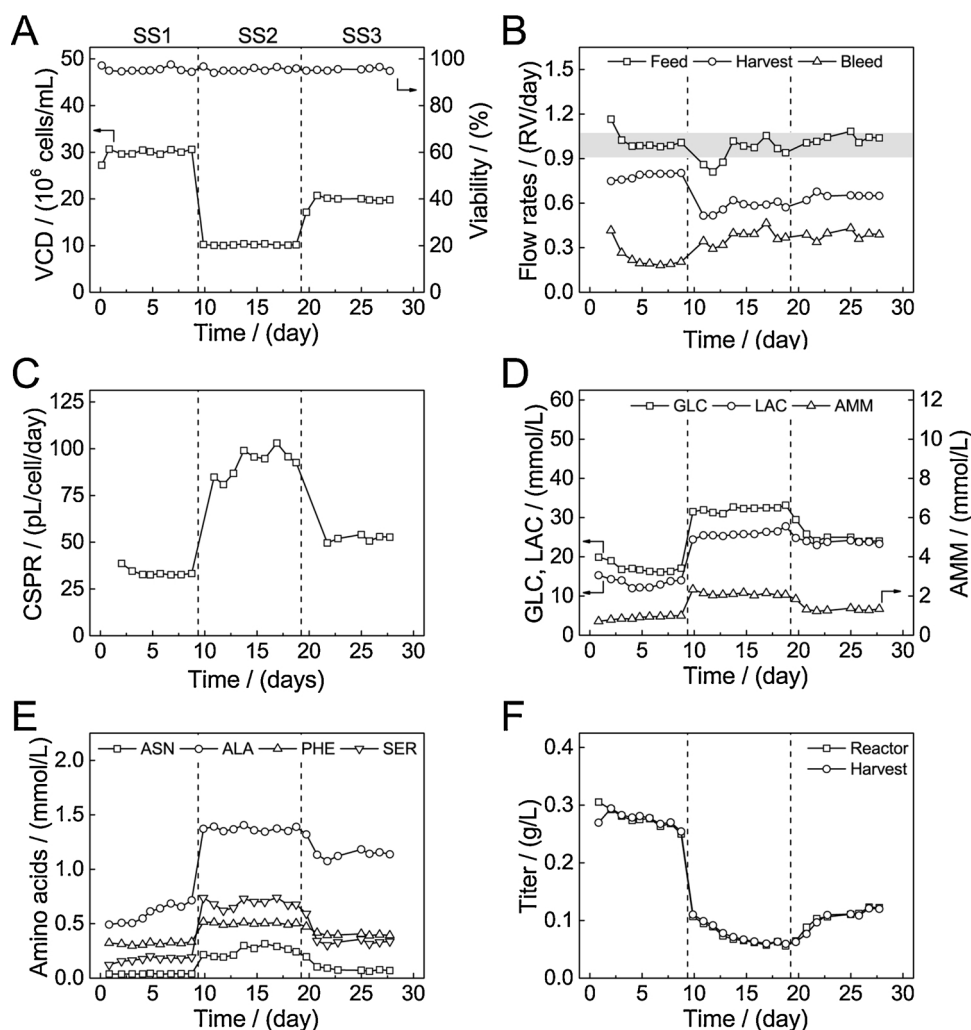


Fig. 2. Step 1 A: Sequence of three steady states with VCDs of 30, 10, and 20 $\times 10^6$ cells/mL and constant perfusion rate of 1 RV/day: A) VCD and viability, B) Flow rates, C) CSPR, D) GLC, LAC, AMM concentrations, E) Concentrations of some amino acids (asparagine (ASN), alanine (ALA), phenylalanine (PHE), serine (SER), and F) Product titer in reactor and harvest.

In the first experiment (Step 1A), different cell specific perfusion rates were targeted. The VCD was varied sequentially from 30 to 10, and then to 20×10^6 cells/mL at a constant perfusion rate of 1.0 RV/day. The VCD set points were pre-tested in orbitally shaken tube experiments [37]. The next experiment (Step 1B) was performed at a constant VCD of 20×10^6 cells/mL. The perfusion rate was varied from 1.0 to 0.66 and 2.0 RV/day with a corresponding change in the CSPR. The third experiment (Step 2) was performed at a constant CSPR of 50 pL/cell/day. The VCD was varied sequentially from 20 to 40, and then to 30×10^6 cells/mL, while the perfusion rate was adjusted accordingly.

In all experiments, the perfusion rate was controlled by a gravimetric feedback loop to keep the reactor weight constant. The bleed rate was adjusted by an automated controller using an online biomass sensor (Aber Instruments Ltd., UK) in order to keep the VCD at the desired set point. Finally, the harvest flow rate was kept constant at a given set point. Daily on-demand addition of antifoam agent (Gibco FoamAway™, Life Technologies) prevented foam build-up.

2.2. Analytics

2.2.1. Offline measurements

Bioreactor samples were withdrawn at least once a day. Cell density and viability were analyzed using trypan blue exclusion method (Cedex XS, Roche Diagnostics, Switzerland). Glucose (GLC) and lactate (LAC)

concentrations were measured using the SuperGL compact blood analyzer (Hitado, Germany). Ammonia (AMM) was determined using the CedexBio Analyzer (Roche Diagnostics, Switzerland). All samples were analyzed for mAb titer, amino acid (AA) concentrations and product quality. The protein concentrations were determined using Protein A affinity chromatography. AA concentrations were measured using reverse phase chromatography employing ZORBAX Eclipse Plus C18 stationary phase columns according to the manufacturer's protocol [38].

2.2.2. Product quality analysis

In a first step, supernatant samples were purified using small-scale affinity columns (PhytipsVR, PhyNexus, San Jose, CA). The analysis of N-linked glycosylation was performed by capillary gel electrophoresis with laser-induced fluorescence detection (CGE-LIF, DNA genetic analyzer 3130XL, Life Technologies, Darmstadt, Germany). The imaged capillary isoelectric focusing method (iCE280 analyzer, ProteinSimple, Santa Clara, CA) was applied for determining charge variant clusters. In total, five main glycan species and five different charge variant clusters (two acidic, one neutral, two basic) were identified.

2.3. Cell specific rates, harvest yield, and volumetric productivity

The cell specific growth, μ was computed according to Eq. (2), while the cell specific consumption/production rates of key metabolites, q_i

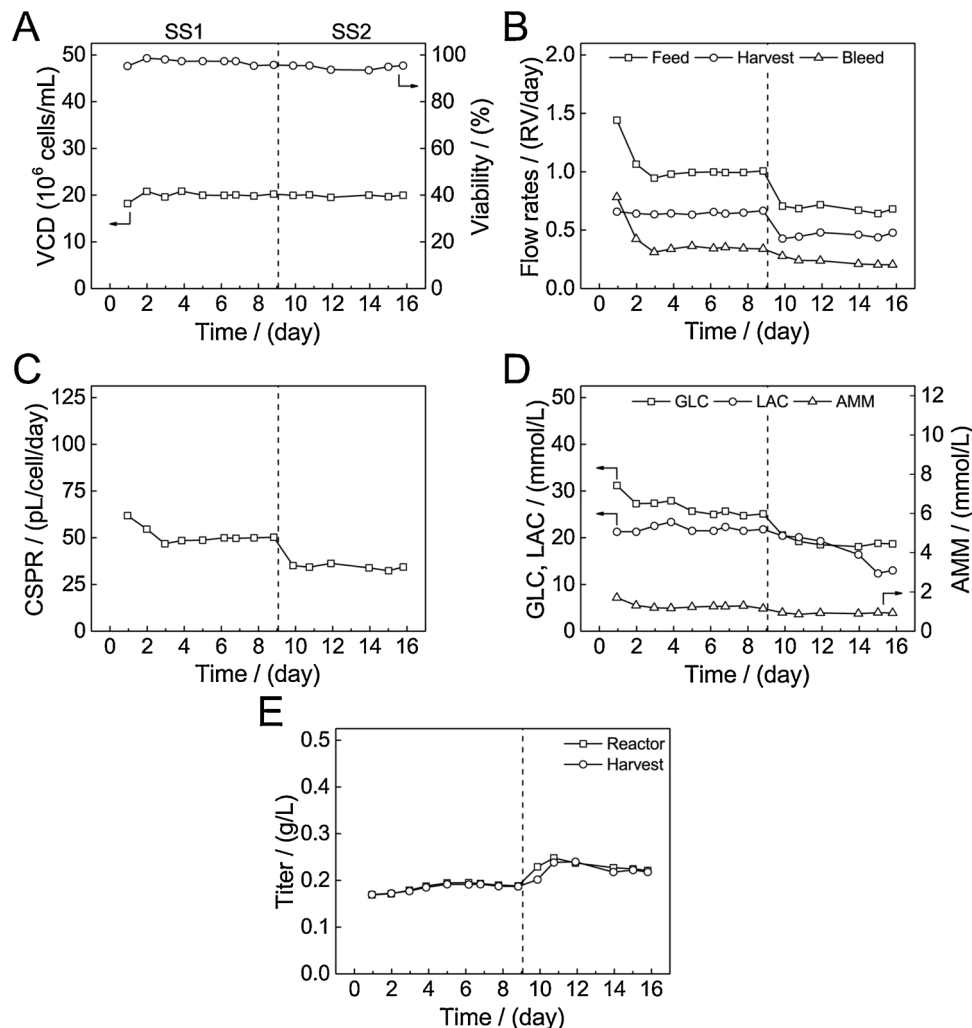


Fig. 3. Step 1 B: Sequence of two steady states at a VCD of 20×10^6 cells/mL with a perfusion rate of 1.0 and 0.66 RV/day: A) VCD and viability, B) Flow rates, C) CSPR, D) GLC, LAC, AMM concentration, E) Product titer in reactor and harvest.

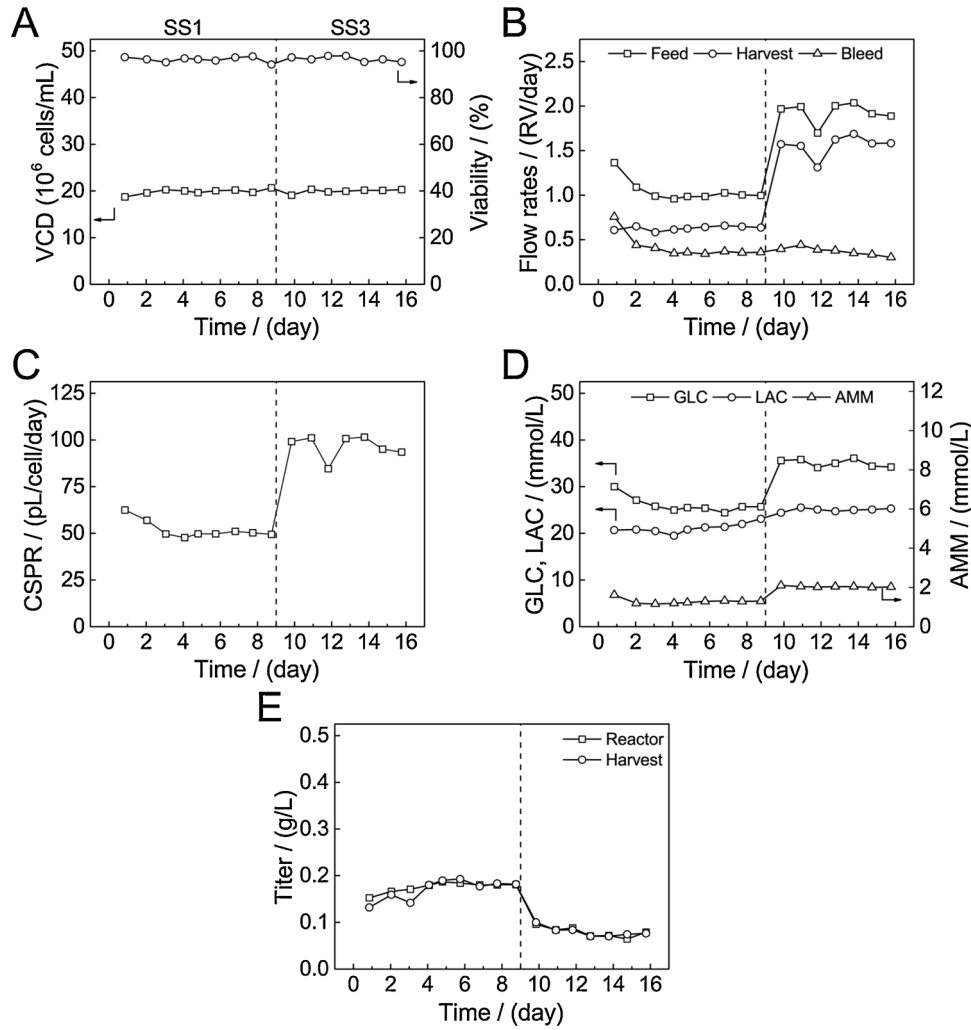


Fig. 4. Step 1 B: Sequence of two steady states at a VCD of 20×10^6 cells/mL with a perfusion rate of 1.0 and 2.0 RV/day: A) VCD and viability, B) Flow rates, C) CSPR, D) GLC, LAC, AMM concentration, E) Product titer in reactor and harvest.

and the harvest yield, and the volumetric productivity at steady state were calculated as follows:

$$q_i = ((c_i^{in} - \bar{c}_i)(H + B)) \frac{1}{VCD} \quad (4)$$

$$Yield_H = \frac{H}{P} \quad (5)$$

$$VP = \frac{H X_V q_{mAb}}{P} = c_{Titer} H \quad (6)$$

Where X_V , is the viable cell density and $\bar{X}_V$ its daily average; t , time, P , H and B the daily perfusion/feed, harvest, and bleed rate normalized to the reactor volume, respectively, c_i the concentration of the specific metabolite i and $\bar{c}_i$, its daily average; c_i^{in} the concentration of the specific metabolite in the fresh media. The harvest yield, $Yield_H$, describes the fraction of the overall volumetric throughput that leaves the reactor with the harvest stream. A detailed list of the used equations is given in the supplementary information (See Section Equations).

3. Results

3.1. Cell culture performance

3.1.1. Step 1A: $CSPR_{min}$ estimation at constant perfusion rate

The first step of the design procedure aims at determining the minimum CSPR value following the *push-to-high* approach. In

particular, we investigated a long-term perfusion run targeting three different viable cell densities (30 , 10 , and 20×10^6 cells/mL) at a constant perfusion rate of 1 RV/day and evaluated the effect on cellular growth and metabolism at steady state. The values of VCD and viability, the flowrates, metabolite concentrations, various amino acid concentrations, and mAb titer are shown in Fig. 2 as a function of time. Each of the VCD set points was maintained constant at high viability for more than a week to guarantee the establishment of steady-state conditions (Fig. 2A). The harvest rate was adjusted according to the VCD as follows: 0.80 RV/day at 30×10^6 cells/mL, 0.60 RV/day at 10×10^6 cells/mL, and 0.65 RV/day at 20×10^6 cells/mL. The perfusion rate was kept stable during all three steady states (SS1: 1.00 ± 0.02 RV/day, SS2: 0.99 ± 0.04 RV/day, SS3: 1.03 ± 0.03 RV/day) close to the nominal set point of 1.00 RV/day (Fig. 2B). The resulting cell specific perfusion rates, given by Eq. (1), are 32.9 ± 0.3 pL/cell/day, 96.3 ± 3.5 pL/cell/day, and 52.5 ± 1.2 pL/cell/day (Fig. 2C), respectively. The cellular growth decreased with the CSPR. Between 10 and 20×10^6 cells/mL, the specific growth rate varied slightly between 0.40 day^{-1} and 0.38 day^{-1} . At 30×10^6 cells/mL, the cell specific growth rate decreased to 0.21 day^{-1} .

The change of the VCD set point and, thus, the CSPR resulted in differences in the metabolite (GLC, LAC) and AA concentrations (Fig. 2D and E). The steady-state levels of various metabolites (GLC, LAC, and various AAs) were lower with higher cell densities. The lower levels of LAC and AMM (See supplementary section Production of

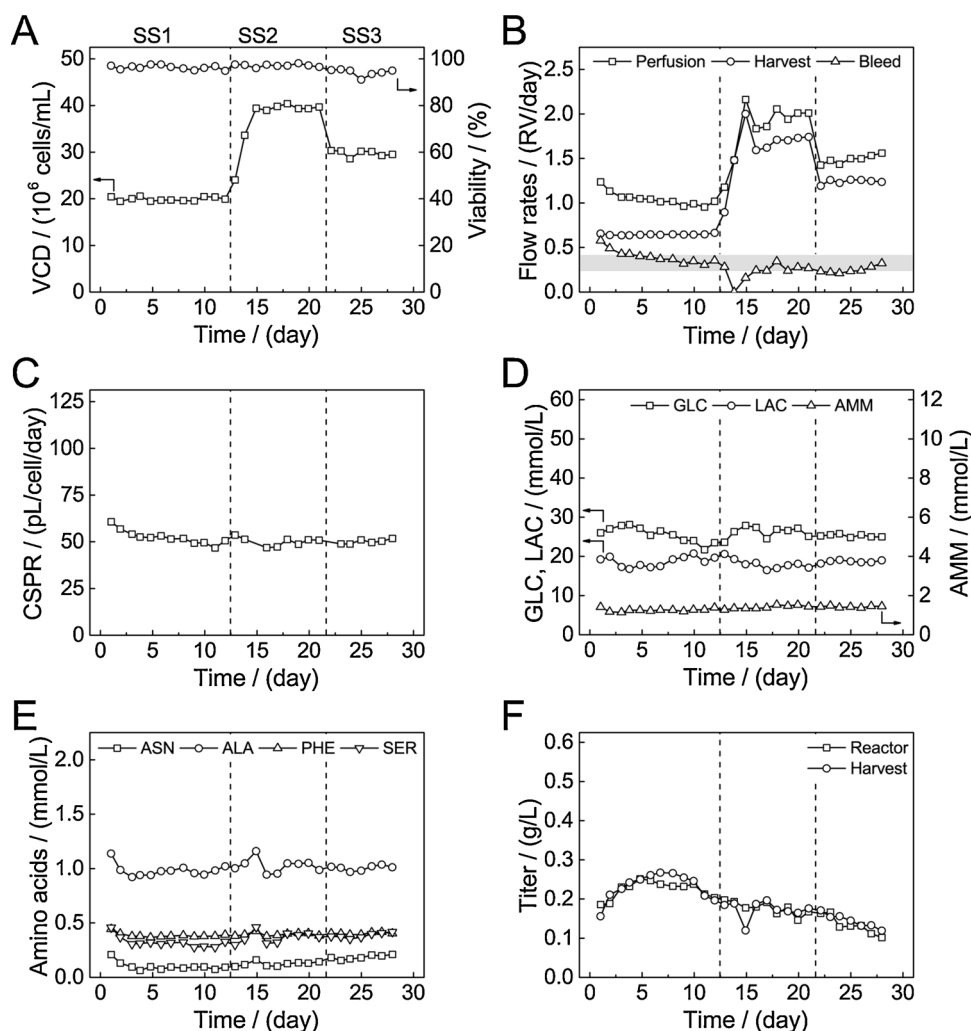


Fig. 5. Step 2: Sequence of three steady states with VCDs of 20, 40, and 30×10^6 cells/mL and constant cell specific perfusion rate: A) VCD and viability, B) Flow rates, C) CSPR, D) GLC, LAC, AMM concentrations, E) Concentrations of some amino acids (asparagine (ASN), alanine (ALA), phenylalanine (PHE), serine (SER)), and F) Product titer in reactor and harvest.

ammonia for fixed medium composition) indicate more dramatic metabolic shifts. When the molar ratio of produced lactate per consumed glucose decreases, this typically represents a shift towards a more efficient metabolic state, which is often observed with lower nutrient availability [39,40].

The remaining AA concentrations are shown as a function of time in **Figure S2** in the supplementary material. It is seen that the flow rates and the metabolites took 1–2 days to achieve steady-state conditions after each set point change. This corresponds to about two times the average reactor residence time, which is the typical duration of the flow dynamic transient in stirred vessels. The detailed experimental data are reported in the supplementary material (**Tables S1** and **S2**).

Measured IgG titers followed the viable cell density profile (**Fig. 1F**), but also indicated a long-term decrease in cellular productivity, probably due to some cell aging / cell line instability process, as observed in previous studies using the same [10] or different cell lines [7].

It is worth pointing out that at the CSPR of 32.9 ± 0.3 pL/cell/day, the bleed and nutrient levels were very low. The corresponding bleed rate of 0.21 day^{-1} indicates that we did not push the CSPR down to the limit, but on the other hand, we considered it to be sufficiently low and we did not want to risk the culture's stability. Therefore, for illustrative purposes, we took this value as the CSPR_{min}.

3.1.2. Step 1B: CSPR_{min} estimation at constant VCD

This experiment represents an alternative to the previous one to determine the CSPR_{min} value, based on the *push-to-low* approach [31]. In particular, the VCD was kept constant at 20×10^6 cells/mL and the effect of varying the perfusion rate was investigated. Three levels of the perfusion rate were considered: 1.00, 0.66, and 2.00 RV/day (for SS1, SS2, SS3), leading to CSPRs of 50, 33, and 100 pL/cell/day, respectively. However, in this case the three steady states were not reached sequentially as in the previous case. Instead, two experiments were performed by first reaching the steady state SS1, and then changing the perfusion rate to reach either SS2 or SS3. The corresponding experimental results are shown in **Figs. 3** and **4**, respectively. It is seen that the VCD value remained stable at high viability in both experimental runs (**Figs. 3** and **4A**). Lowering the perfusion rate / CSPR led to a decrease in the bleed rate as well as in GLC, LAC, and AMM concentrations, but increased mAb titers (**Fig. 3B–E**). On the other hand, doubling the perfusion rate from 1.0 to 2.0 RV/day, slightly increased the bleed rate and led to higher levels of GLC, LAC, and AMM, while the titer decreased from 0.18 to 0.07 g/L (**Fig. 4B–E**). The repetition of SS1 at 50 pL/cell/day was performed to show the reproducibility of the experimental procedure.

It is worth noting that the smallest value of CSPR found in this Step 1B supports the identified CSPR_{min} of 33 pL/cell/day with comparable bleed rates of 0.20 day^{-1} .

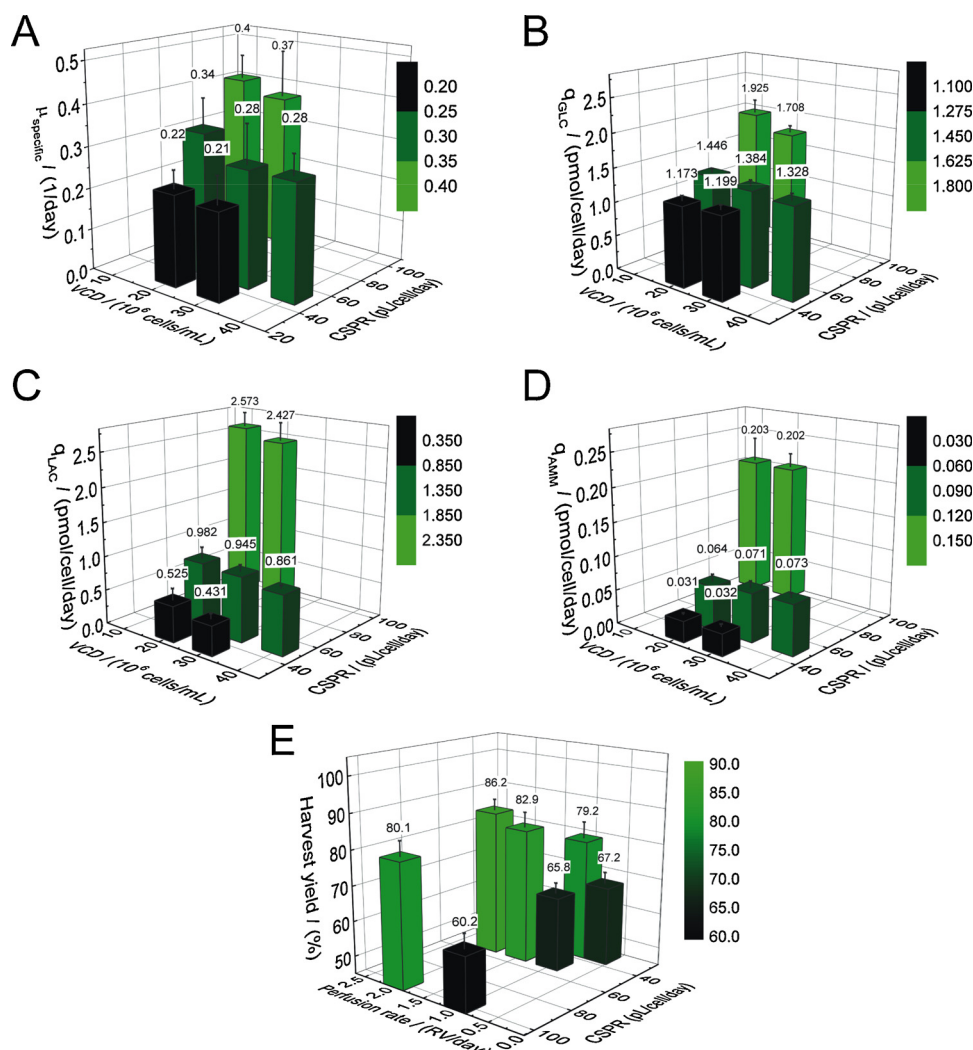


Fig. 6. Steady-state values of specific rates as a function of the VCD and CSPR: A) Cell specific growth rate, B) Cell specific glucose consumption rate, C) Cell specific lactate production, and D) Cell specific ammonia production. E) Harvest yield as a function of P and CSPR. The steady state values are calculated as average of the last 5 days. Error bars represent two standard deviations.

3.1.3. Step 2: operation at constant CSPR

The second step of the design procedure consists of long-term perfusion runs at constant CSPR. For this purpose we chose the mid-point CSPR of 50 pL/cell/day and three different VCDs (20, 40, and 30 $\times 10^6$ cells/mL, corresponding to SS1, SS2, and SS3) to evaluate the effect of varying VCD and perfusion rate on process performance for a fixed CSPR. The values of VCD and viability, flowrates, metabolites, various amino acid concentrations, and mAb titer are shown in Fig. 5 as a function of time. Each of the three different VCD set points was maintained constant at high viability for more than a week (Fig. 5A). The perfusion rate had to be adjusted, by properly changing the set point of the harvest pump after each switch to a new VCD set point, to maintain a constant CSPR according to Eq. (3). The consistent growth resulted in unchanged bleed rates of around 0.30 ± 0.05 RV/day as highlighted by the grey area representing the average bleed in Fig. 5B. This enabled the control of the perfusion rate (SS1: 0.98 ± 0.03 RV/day, SS2: 2.01 ± 0.04 RV/day, and SS3: 1.52 ± 0.03 RV/day) by only adjusting the harvest rate. The transition from the first steady state to the next one was done by a step-by-step increase of the harvest flowrate until the desired VCD was reached in order to maintain the CSPR constant. Thus, the CSPRs was kept stable at 49.6 ± 1.7 , 49.8 ± 1.6 , and 50.3 ± 1.0 pL/cell/day for the three operating points as shown in Fig. 5C. The concentrations of GLC, LAC, AMM, and various AAs were constant over time and independent of the VCD value as shown in

Fig. 5D and E. The remaining AA concentrations as a function of time are shown in Figure S3 in the supplementary material.

The mAb concentration decreased over time (Fig. 5F). The measured peak concentration was 0.25 g/L on day 6. With ongoing cultivation, the measured concentrations decreased steadily and flattened out at the end of the culture to 0.12 g/L. The drop in specific productivity was observed in all runs as shown in the supplementary Figure S4.

3.1.4. The two-step design procedure

Let us now consider all the data collected in the three experiments above. Fig. 6A–D summarize the steady-state cell specific rates (growth rate, metabolic uptake and consumption rates) as a function of CSPR and VCD in the ranges of 33–100 pL/cell/day and $10\text{--}40 \times 10^6$ cells/mL, respectively. It is seen that, for a given CSPR value, cellular growth and metabolism do not significantly change at different VCDs, while they are strongly affected by altering the CSPR value, independent of the sequence of steady states that were performed. In particular, as expected, decreasing the CSPR slowed down cellular growth as well as metabolite in- and output rates and vice versa. These findings are in agreement with previous studies [21,41] and underline the importance of the CSPR as a key control parameter in perfusion cultures [31,32].

It is worth noting at this point, that the cell line under consideration exhibited a consistent drop in cell specific productivity and therefore

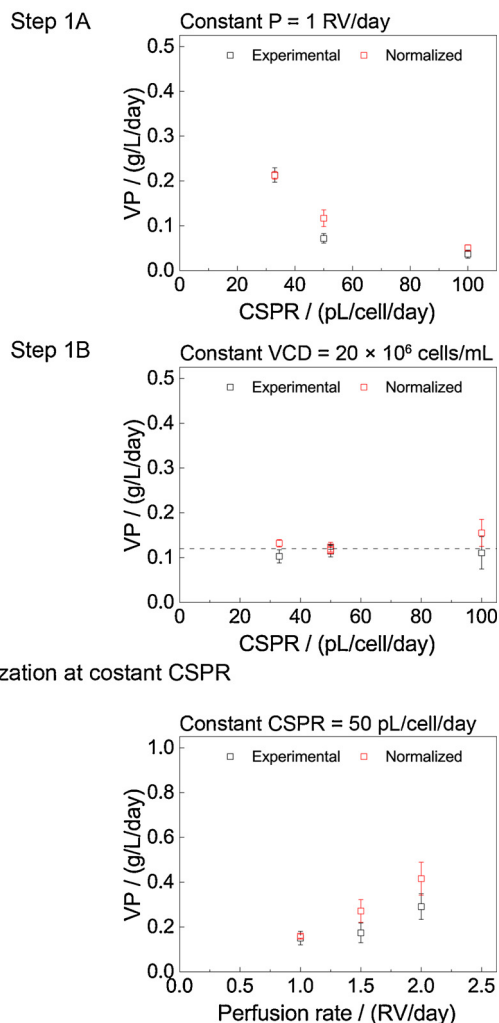
Step 1: Estimation of $CSPR_{min}$ 

Fig. 7. Experimental (□) and normalized (□) steady-state value of the volumetric productivity for each step of the optimization procedure. In Step 1A, the volumetric productivity increases with decreasing CSPR due to the higher titers at constant perfusion rate. In Step 1B, the volumetric productivity remains constant as both CSPR and volumetric throughput are varied, and in Step 1A, the volumetric productivity increases with increasing P at constant CSPR.

neither the cell specific productivity nor the medium consumption were comparable due to their dependence on cultivation time..

The same experimental runs are considered in Fig. 6E in terms of the harvest yield as a function of the perfusion rate and the CSPR. The harvest yield only describes the volumetric harvest fraction without taking the cell specific productivity/protein concentration into account and is thus independent of the cultivation time. It was observed that by reducing the CSPR, since the cellular growth and consequently the bleed rate decrease, the harvest yield increased while keeping the perfusion rate constant. For example in Step 1A, one can decrease the CSPR from 100 down to 33 pL/cell/day while increasing the yield from 60% to 80%. Once a suitable minimum CSPR is identified, the culture operating conditions can be further improved while keeping the CSPR constant. Accordingly, in Step 2 we increased the VCD and the perfusion rate in order to maintain a constant CSPR of 50 pL/cell/day. This entailed increasing the harvest rate from 0.7 to 1.2 and then to 1.7 RV/day while the bleed remained constant in order to operate at 20, 30, and 40 × 10⁶ cells/mL, respectively. This allowed elevating the harvest yield from 65% to 85%, calculated according to Eq. (4).

In addition, we want to consider the volumetric productivity within

the different steady states of the three steps. The steady-state values of the volumetric productivity are presented as a function of the CSPR (Step 1) and of the perfusion rate (Step 2) in Fig. 7. Due to the observed productivity drop, we also calculated a normalized volumetric productivity. The normalized productivity was calculated by dividing the daily productivity by the average drop (%) observed in all runs reported in the supplementary Figure S4. We also show both volumetric productivities as a function of time in the supplementary Figure S5. At constant perfusion rate (Step 1A), decreasing the CSPR results in higher volumetric productivities due to improved harvest yields and higher cell densities (Eq. 6). At constant VCD, but changing perfusion rates (Step 1B), the volumetric productivity remains rather constant as both the CSPR and the volumetric throughput are lowered at the same time. At constant CSPR, however, the operation at higher VCD and perfusion rate set points is beneficial enabling higher volumetric productivities, even with respect to the observed productivity drop.

3.2. Product quality

In the experimental runs described above, the time evolution of N-linked glycosylation and charge isoforms was also investigated. Five different glycan moieties were measured: high mannose structures (HM), the fucosylated FA2G0 form, which is further processed to the one sided FA2G1, and then to double sided FA2G2 galactosylated forms, and finally the sialylated glycan moieties (SIA). A lumped reaction network connecting these species is reported in the supplementary information (Figure S6). Considering the protein's different charge isoforms, five differently charged isoforms were identified: two acidic charge variants CV1 and CV2, the neutral and main charge variant CV3, and two basic charge variants CV4 and CV5.

3.2.1. Step 1 A: $CSPR_{min}$ estimation at constant perfusion rate

3.2.1.1. Charge variants. Fig. 8A shows the different charge isoforms for the long-term culture targeting different CSPRs at constant perfusion rate as a function of time. The neutral charge variant CV3 and the acidic charge variant CV2 were the two dominant species throughout the entire experiment, followed by the CV1 and CV4 isoforms. The fraction of CV5 remained below 5% for all operating points. Right after each set point change, the charge isoform distribution changed slightly during the first two days and then settled to a constant pattern after reaching steady-state conditions. Comparing the different steady states, the charge variant distributions showed only minor variations. Average fractions of the main charge variants changed from 66% of CV3 and 16% of CV2 during SS1, to 68% of CV3 and 14% of CV2 during SS2, and finally to 71% and 15% of CV3 and CV2, respectively, during SS3. From these results, no direct correlation between CSPR and resulting charge variant distribution could be established.

3.2.1.2. N-linked glycosylation. Two glycan moieties dominated the glycosylation pattern of the produced mAb: the FA2G0 and the FA2G1 forms. All the other structures (HM, FA2G2, and SIA) were below 5% as shown in Fig. 8B. The time profile of the glycans revealed a transition phase of 3–4 days at the beginning of each transient towards a distribution, which remained stable at steady state. It appeared that different steady-state glycan distributions were not exactly the same, but indeed rather similar. The average fraction of FA2G0 varied from 75% to 76% to 78% between the three consecutive steady states, while the FA2G1 fraction maintained around 18% between SS1 and SS2 and then changed to 16% in SS3. The ratio of the FA2G0 and the FA2G1 forms (also known as G0:G1 ratio) remained in the range of 4.2 to 5.0 for the three different conditions. Considering these low variations between the different CSPRs, no strong correlation between glycosylation and CSPR could be determined. The reaction converting FA2G0 to FA2G1 is catalyzed by galactosyltransferase (GalT). Among others, ammonia and the inner pH in the golgi apparatus can strongly affect galactosylation [35,42,43]. In this experiment, the ammonia

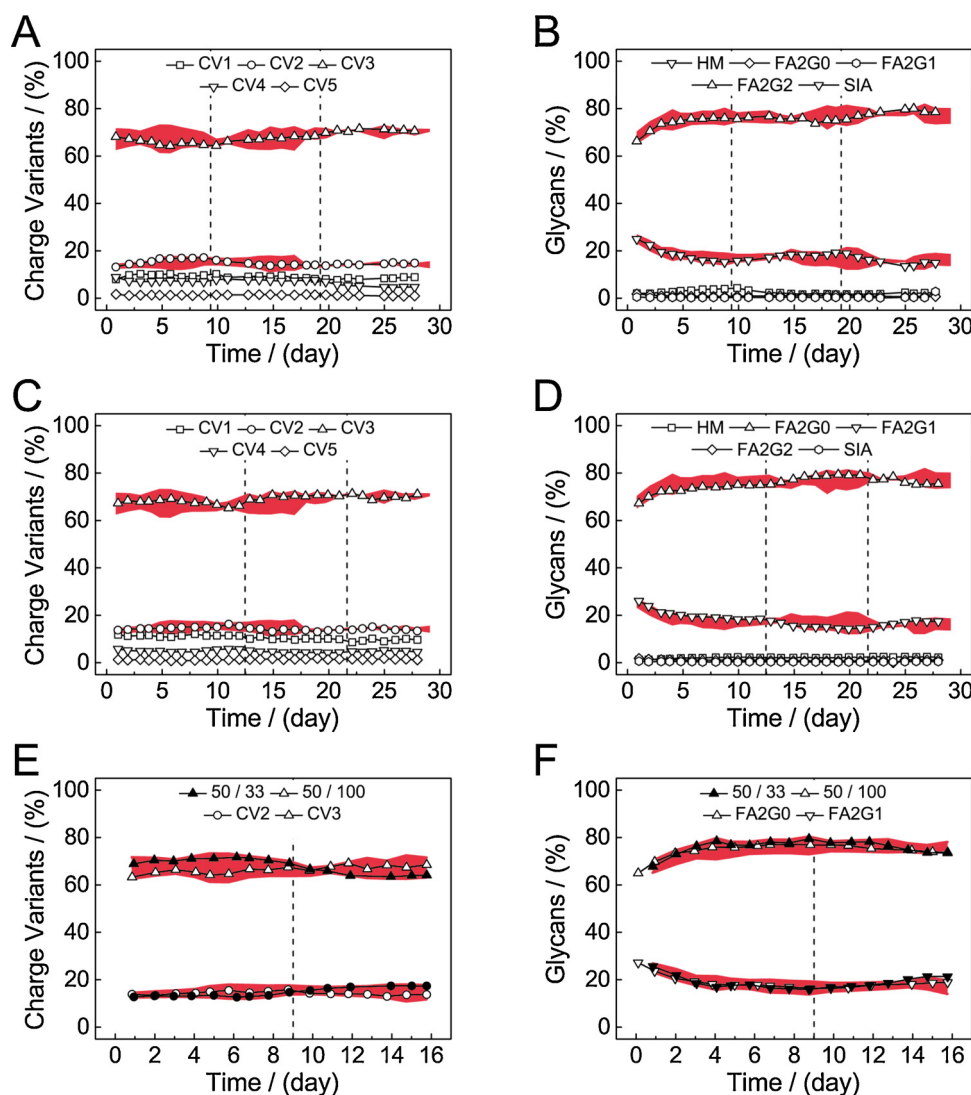


Fig. 8. Charge variants (A, C, E) and main glycoforms (B, D, F) as a function of time in the different experimental runs: A & B) Step 1 A: constant perfusion rate, C & D) Step 2: constant cell specific perfusion rate, E & F) Step 1 B: constant viable cell density. The red area represents the average value and two standard deviations of all experiments over time (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

concentration did not exceed 2 mM during all steady states (Fig. 2D) leading to a similar galactosylation efficiency. For the tested range, these results indicate that at a fixed medium composition and low ammonia accumulation the glycosylation of a protein remains constant at a stable operation, even if the viable cell density levels are varied.

3.2.2. Step 2: operation at constant CSPR

3.2.2.1. Charge variants. The time profiles of the different charge variants at constant CSPR are presented in Fig. 8C. The dominant isoforms were again the neutral CV3 and the acidic CV2 forms. As in the previous case, during each steady state the charge variant distribution remained rather constant and showed only minor variations between the different operating points. In particular, the average fraction of the main charge variant CV3 slightly varied from 67% to 71% to 70%, while CV2 stayed within a range of 14–15% in the three steady states. It seems that with a higher perfusion rate ($SS2 > SS3 > SS1$), and consequently shorter residence time, the fraction of CV3 slightly increases, while CV2 and the other fractions decrease. This would be consistent with a posttranslational modification of the main variant to produce the others. However, the results are similar to that at constant perfusion rate and the variations are within the error of detection.

3.2.2.2. N-linked glycosylation. The glycan patterns are presented in Fig. 8D as a function of time. The dominant glycan moieties were once again the FA2G0 and the FA2G1 forms. After each set point change, the glycan forms changed for about 3–4 days before reaching steady state conditions. The average fraction of FA2G0 varied from 75% to 79% to 76%, while the FA2G1 fraction was 19%, 15%, and 17%. The G0:G1 ratio remained in the range between 4.0 and 5.4. The very small variations between each steady state are consistent with the constant ammonia levels between 1.2–1.5 mM as shown in Fig. 5D.

3.2.3. Overall product quality considerations

The rather small variations between charge isoforms and glycan patterns observed in the experiments at constant perfusion rate (Step 1 A) and constant CSPR (Step 2) discussed above, were also found in the constant VCD experiments (Step 1 B) as shown in Fig. 8E and F for the dominant moieties. In Fig. 8, the red areas represent the average of the charge variant and glycan distributions of all runs. Comparing the average distributions with those obtained in the different experiments reveals that both product quality attributes remained similar with variations of $\pm 5\%$ independent of the operating conditions. Minor variations were observed, especially right after a set point change during transient conditions. The charge variants and glycan moieties

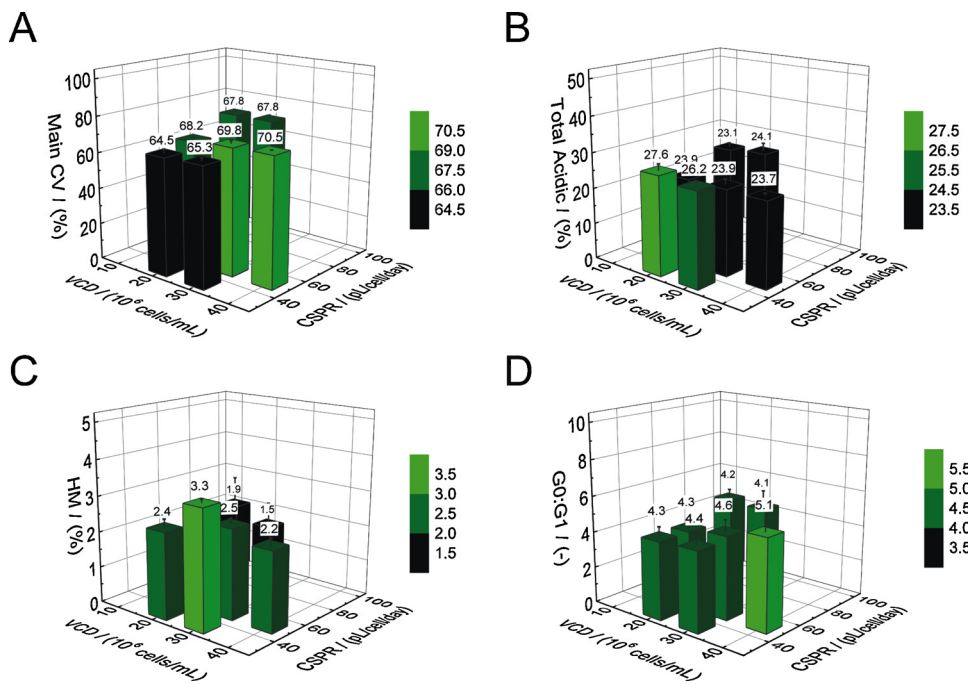


Fig. 9. Steady-state values of product quality attributes as a function of VCD and CSPR: A) Average fraction of the main charge isoform CV3, B) Average total fraction of acidic charge variants (CV1 + CV2), C) Average fraction of high mannose glycan moieties, D) Average ratio of FA2G0 and FA2G1. The steady state value is calculated as an average of the last 5 days. Error bars represent two standard deviations.

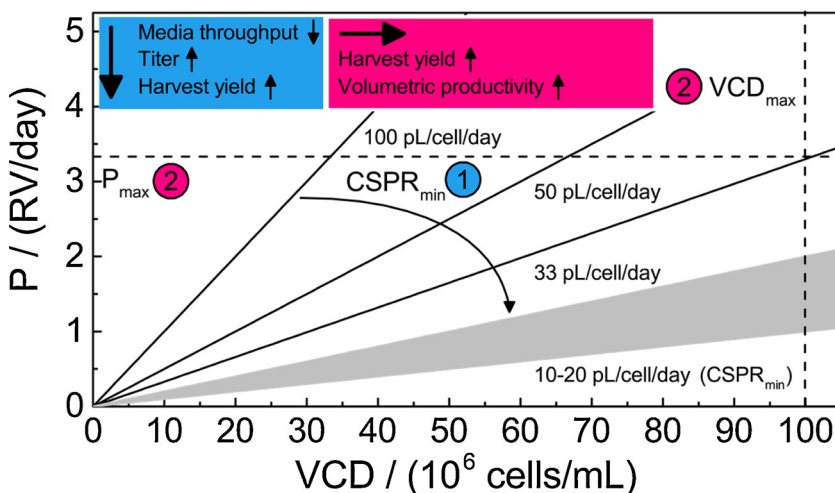


Fig. 10. Two-step procedure for defining optimal operating conditions for perfusion bioreactors: Step 1 includes defining a convenient CSPR (i.e. the CSPR_{min}) by either applying experiments at constant perfusion rate (Step 1 A) or constant VCD (Step 1 B), but decreasing CSPR. In the second step, optimal operating conditions are defined by going along the chosen CSPR with increasing VCD and P until operating boundaries are reached (Step 2).

are presented as a function of time for the Step 1B experiments in the supplementary material (Figures S7 and S8).

The product quality data regarding steady-state values are summarized in Fig. 9. The main charge variant (CV 3) and the total acidic charge isoforms (Fig. 9A and B) show a minor increase in acidic forms as the CSPR decreases. For glycosylation, a narrow range of the G0:G1 ratio and the high mannose moieties is observed for the different experiments (Fig. 9C and D). Within the investigated ranges of cell densities ($10\text{--}40 \times 10^6$ cells/mL) and perfusion rates ($0.66\text{--}2$ RV/day) for the given medium composition only minor variation in the range of $\pm 5\%$ were observed.

4. Discussion

In this work we develop a two-step procedure for designing the operating conditions of a perfusion bioreactor (Fig. 10). Reducing the CSPR at a fixed perfusion rate by a stepwise increase of the VCD (Step 1 A) or at a fixed VCD by a stepwise decrease of the perfusion rate (Step 1 B) results in decreased cellular growth rates and, therefore, lower bleed rates. Additionally, lower by-product levels (AMM and LAC) indicate a

metabolic shift towards a more efficient metabolic state [40,41]. It is confirmed that both strategies, *push-to-high* and *push-to-low*, result in a similar CSPR_{min} that we defined to be 33 pL/cell/day in this case, which is the designated first step of the design procedure. Indeed, we did not push the CSPR to the limit in this work to preserve the culture stability.

For a given medium, operation at CSPR_{min} defines the best operating point of the cell culture [31], since it guarantees low product loss in the bleed stream, as well as large productivities and small media consumption. In the second step, the operating conditions in terms of perfusion rate and VCD have to be defined (Step 2) while keeping their ratio equal to the CSPR_{min}. Our study shows that operating at constant CSPR, but increased VCDs and perfusion rates leads to a further improvement in the bioreactor performance, also in agreement with recent results of Lin et al. [34].

In terms of product quality attributes (e.g., charge isoforms, N-linked glycosylation), these exhibit minor fluctuations during steady-state operation. On the other hand, at least for the conditions investigated in this work, the observed product quality attributes were found to be substantially independent of CSPR, VCD and perfusion rate. It is worth noting that increased perfusion rates are favorable in case of

a fragile protein reducing the average residence time of the protein and thus the time for undesired degradation or modifications reactions. More significant changes in the product quality can be induced through different actions, such as adding feed supplements, or by adjusting the bioreactor conditions (temperature, pH, osmolality), as shown for the glycosylation pattern in other studies [35,44–46].

An important observation refers to the duration of the transient needed to achieve steady-state conditions. With respect to metabolites, cellular metabolism and growth, flow rates and charge isoform patterns took about 1–2 days. The glycoform distribution, in contrast, needed up to 4 days to reach steady-state levels. The initial conditions in the reactor are of course relevant with respect to the overall dynamics. Karst et al. have reported an initial transition phase of about 3 days for metabolites and reactor hydrodynamics, and about 7 days for glycan patterns to reach steady state [35,36]. In this study the time to steady state was shorter. The difference is that Karst et al. used two different medium compositions between the seeding bioreactor and the N-stage bioreactor, while in this case the perfusion medium composition remained unchanged. The change in medium represented a more drastic difference in the metabolite environment between the N-1 and the N-stage resulting in a prolonged transient phase for product quality attributes, especially in the case of glycosylation.

We observed that the cell line used in this work exhibited a drop in the specific mAb productivity over time as a result of intrinsic cell line instability and aging. For the same cell line, this phenomenon was observed by Karst et al. in several experiments [10,28]. Bertrand et al. [47] concluded that intrinsic cell line instability was the result of epigenetic promoter silencing for the cell line under consideration. Considering the cell specific productivity, the used cell line diverges from the ideal behavior as the decline in productivity is time dependent. This could be indeed verified within the three experiments performed in **Step 1** of the procedure, where independent of the sequence of steady states, similar productivity drops were observed. An even better verification strategy to account for this phenomenon during the optimization procedure would be to test to test the cell stability within a long-term control experiment without changing either CSPR, or VCD, or P. In our case, we compared and calculated the average productivity drop observed in all the four experiments as a function of time, and could indeed identify a similar trend in all experiments (Supplementary **Figure S4**), which we also took into account when calculating the normalized volumetric productivity in **Fig. 7**.

5. Conclusions

The optimization of a perfusion bioreactor operation is a complex process which can be performed following the guidelines illustrated in this work. In a preliminary phase, some performance criteria and operating boundaries have to be established. For a given medium, these include minimum cell growth, acceptable byproduct levels, desired product quality, the maximum supportable VCD, VCD_{max} (defined by the bioreactor set up, cell retention device, and operational limitations in terms of oxygen supply and CO_2 removal), and a maximum achievable perfusion rate, P_{max} (in terms of medium volumes and economic considerations.) Based on these criteria, the most convenient CSPR, i.e. the minimum sustainable $CSPR_{min}$, can be found through either experiment Step 1 A or Step 1 B, which involves a sequence of steady-state runs with either constant perfusion rate or VCD, but decreasing CSPR. Once a convenient CSPR has been found, the optimal process parameters can be obtained through experiment Step 2, that is going along the $CSPR_{min}$ with increasing VCD and P until either VCD_{max} or P_{max} are reached. This procedure is illustrated in **Fig. 10**. Operating at higher VCDs and consequently higher perfusion rates will decrease the residence time of the product (especially beneficial for less stable proteins) and increase the harvest yield. At optimum operating conditions ($CSPR_{min}$, VCD_{max} , and/or P_{max}), product quality can be further fine-tuned in an additional step, for example by adding proper supplements

in the reactor feed [35]. The data reported in this work are of course specific for a given expression system and target protein, however, the approach is rather general and can be transported to other systems.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.bej.2019.107295>.

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